

[A] Single-Sequence Prediction of Tertiary Protein Structure

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To address the computational overhead and evolutionary data dependencies of MSA-based models like AlphaFold, this project develops a single-sequence methodology that learns the latent grammar of proteins via a lightweight Transformer architecture. Operating on a coarse-grained C_α representation, the model branches early into a deep 1D sequence track and a shallow 2D pairwise track. These converge via spatial attention pooling prior to 3D prediction and optimizer refinement, reaching a mean TM-Score of 0.31. This design avoids the computational bottlenecks of evoformers and deep 2D tracks while still leveraging 2D spatial awareness to predict accurate global topology directly from a single sequence.

Project Topic: Prediction of protein structure from sequence

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I. INTRODUCTION

Determining the three-dimensional topology of a protein strictly from its one-dimensional amino acid sequence is a primary computational challenge in modern biophysics [1]. For decades, structural biology relied on experimental methods like X-ray crystallography, which provide ground-truth data but are bottlenecked by high costs and an inability to crystalize highly dynamic proteins. The introduction of deep learning frameworks, such as AlphaFold2, revolutionized this domain by extracting co-evolutionary signals from Multiple Sequence Alignments (MSAs) to achieve near-experimental accuracy [2].

However, MSA-dependent architectures fundamentally struggle with orphan proteins, rapidly mutating viral targets, and synthetic designed sequences that lack evolutionary homologs [1]. To circumvent this dependency and the massive computational overhead of full-atom modeling, emerging paradigms advocate for single-sequence methodologies operating over abstract, coarse-grained geometric representations [3]. This project investigates to what extent a coarse-grained C_α model can predict tertiary structure solely from its amino-acid sequence, utilizing a lightweight Transformer backbone combined with a shallow 2D embedding pair projection. Rather than continuously updating a 2D pair representation throughout the network, the architecture branches early, performing standard 1D sequence processing parallel to a pairwise feature product operation used for distogram prediction and final geometric refinement.

II. OVERVIEW

Tab. I evaluates state-of-the-art architectures to justify the proposed early-branching framework.

III. METHOD

The proposed architecture (Fig. 1) translates single-sequence data into 3D Cartesian coordinates while avoiding the computational bottlenecks of deep pairwise representations.

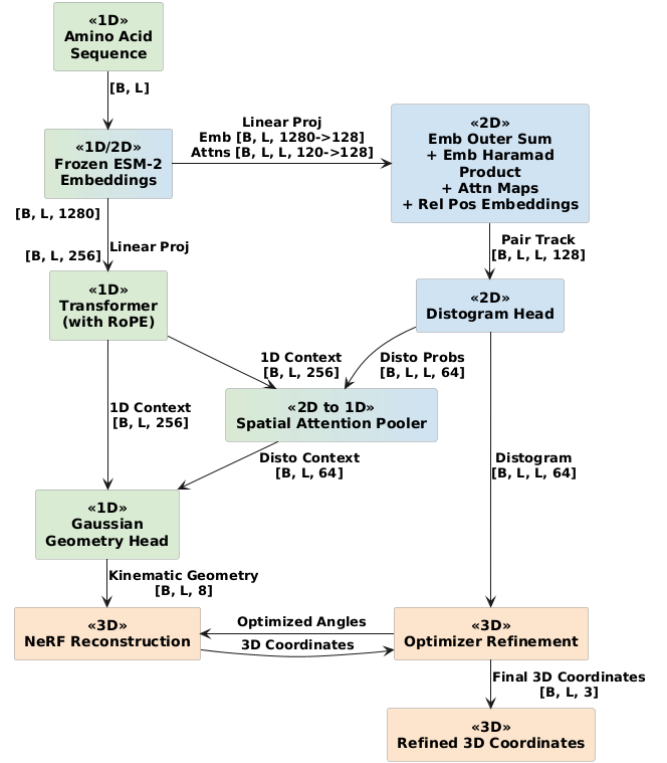


FIG. 1. Single-Sequence Coarse-Grained Transformer Architecture.

A. Model Architecture

Sequence Initialization. A frozen ESM-2 (650M) language model [4] generates a contextualized representation $\mathbf{s}_i$ for every residue i . These embeddings provide rich evolutionary and biochemical priors without requiring Multiple Sequence Alignments.

1D Transformer Track. A 6-layer Transformer processes the 1D sequence features, relying solely on self-attention enhanced with Rotary Positional Embeddings (RoPE) [5] to encode relative sequence distances.

2D Pairwise Track. In parallel to the 1D backbone, a 2D spatial representation is constructed to cap-

TABLE I. **Overview of Architectures and Pipeline Components.** Summarizes the core methodologies and justifies their inclusion or exclusion in the proposed architecture [1, 2].

Method	Use case scenario	Features	Suitable for project?
Iterative Recycling	Deep structural consensus building	Pro: Highly accurate topology resolution. Con: Massive VRAM and inference time penalties.	No. Bypassed for a single-pass topology and L-BFGS solver.
Evoformer & Triangle Updates	Enforcing 2D geometric constraints	Pro: Ensures geometric triangle inequality. Con: $\mathcal{O}(L^3)$ multiplicative updates cause severe bottlenecks.	No. 2D track relies instead on spatial pooling and correlation.
Invariant Point Attn. (IPA)	Direct 3D coordinate attention	Pro: Global SE(3) spatial invariance. Con: Computationally heavy compared to 1D attention.	No. Replaced by 1D Transformers and NeRF conversion.
Frame Aligned Point Error (FAPE)	Local frame-based 3D evaluation	Pro: Aligns independent domain frames perfectly. Con: Highly dense, expensive coordinate calculations.	No. Replaced by a faster sliding-window Distance RMSD.
Sliding-Window 3D dRMSD Loss	Continuous localized 3D evaluation	Pro: Optimizes 3D coordinates natively without triggering catastrophic lever-arm explosions.	Yes. Provides safe, localized geometric gradients.
Early-Branching 1D/2D Topology	Independent spatial and sequence modeling	Pro: Decouples 1D backbone from 2D Distogram, keeping the network fast until late-stage pooling.	Yes. Central to the proposed lightweight architecture.
Rotary Positional Encoding (RoPE)	Length-invariant relative sequence	Pro: Models physical sequence distance offsets, avoiding absolute length bias.	Yes. Critical for relative positional awareness.
Gaussian Geometry Head	Explicit kinematic uncertainty estimation	Pro: Predicts expected angles and uses variance ($\log \sigma^2$) to flag rigidity versus hinge uncertainty.	Yes. Provides the critical threshold masks for refinement.
Binned Distogram Head	Probability-Based topology	Pro: Explicit 2D topological representation with a clear confidence expression.	Yes. Provides context to both Geometry Head and Optimizer.
Confidence-Masked L-BFGS	Surgical downstream 3D refinement	Pro: Freezes confident structures while unleashing gradient descent exclusively on uncertain hinges.	Yes. Acts as the primary global topology solver.

ture pairwise residue interactions. This track operates directly on the embeddings extracted from the frozen language model. First, the raw 1D embeddings are independently projected into left and right representations, $\mathbf{h}_i^{\text{left}}$ and $\mathbf{h}_j^{\text{right}}$. The basic pair representation, $\mathbf{z}_{ij}$, is formed via a composite broadcasted interaction, combining an outer sum with a Hadamard product. The product is scaled as not to overpower the other terms and reduce variance:

$$\mathbf{z}_{ij} = \mathbf{h}_i^{\text{left}} + \mathbf{h}_j^{\text{right}} + \frac{\mathbf{h}_i^{\text{left}} \odot \mathbf{h}_j^{\text{right}}}{\sqrt{d_{\text{pair}}}}$$

With d_{pair} being the dimension of the pair track. To further leverage the structural priors inherent in the language model, the raw 2D attention weights from the final layers of ESM-2 are extracted and symmetrized. These attention maps, $\mathbf{A}_{ij}$, are linearly projected into the pair dimension and stabilized via LayerNorm, which is a critical step to rescue the vanishing signal inherent to $1/L$ attention softmax outputs and not letting it be reduced to background noise when compared to the product term:

$$\mathbf{z}_{ij} \leftarrow \mathbf{z}_{ij} + \text{LayerNorm}(\mathbf{W}_A \mathbf{A}_{ij})$$

Next, clamped relative positional embeddings are added to the representation to encode sequence-distance semantics, up to a maximum threshold of ± 32 residues, providing just enough spatial embedding keys for local structure. The completed 2D track is processed by the Distogram Head. This consists of a Multi-Layer Perceptron (MLP) with GELU activations [6] that projects the features into 64 discrete spatial distance bins ranging from 2–22 Å, where 22 Å represents a non-interacting pair. The resulting pre-softmax logits $\mathbf{D}_{ij}$ are symmetrized so

that the predicted distance $i \rightarrow j$ is identical to $j \rightarrow i$:

$$\mathbf{D}_{ij} = \frac{\text{MLP}(\mathbf{z}_{ij}) + \text{MLP}(\mathbf{z}_{ji})}{2}$$

This distogram provides global 2D structural context and serves as the spatial loss target for both training and final refinement.

Spatial Pooling. To reintegrate spatial information into the sequence representation, a Spatial Attention Pooler compresses the 2D structural relationships into a 1D context vector. This mechanism deviates from a standard attention operation primarily by omitting a distinct “Value” projection. Queries are linearly projected from the 1D sequence embeddings ($\mathbf{h}_i$), while Keys are projected from the 2D probabilities ($\mathbf{P}_{ij}$). The resulting attention weights are then used to directly pool the unprojected probability distributions, computing a weighted sum across the sequence dimension j :

$$\mathbf{c}_i = \sum_{j=1}^L \text{Softmax}_j \left(\frac{(\mathbf{W}_Q \mathbf{h}_i) \cdot (\mathbf{W}_K \mathbf{P}_{ij})}{\sqrt{d_{\text{bins}}}} \right) \mathbf{P}_{ij}$$

B. Kinematic Reconstruction

Gaussian Geometry Head. Following the reintegration of the 2D spatial context, the 1D representation is projected into a final Gaussian Geometry Head to explicitly model the epistemic uncertainty of the local protein backbone. For each residue i , it predicts an 8-dimensional output consisting of means (μ) and log-variances ($\log \sigma^2$) for the coarse-grained internal geometries (specifically C_α virtual pseudo-angles):

$$\mathbf{o}_i = [\mu_\theta, \log \sigma_\theta^2, \mu_\tau, \log \sigma_\tau^2, \mu_d, \log \sigma_d^2]$$

The angular means (μ_θ and μ_τ) are output as 2D L_2 -normalized unit vectors representing the sine and cosine components of the backbone bond angles and torsions. The distance mean (μ_d) predicts the virtual separation to the subsequent C_α atom. The log-variances act as learned confidence indicators, distinguishing rigid secondary structures from highly flexible coils. The predicted internal coordinate means ($\mu_\theta, \mu_\tau, \mu_d$) are passed into the Natural Extension of Reference Frame (NeRF) algorithm [7] to iteratively construct the 3D backbone Cartesian coordinates atom-by-atom.

C. Joint Optimization and Loss Functions

The network is trained end-to-end using a joint objective function balancing probabilistic internal geometry, structural chirality, 2D topology, and localized 3D coordinates:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{geom}}(\mathcal{L}_{\text{NLL}} + \lambda_{\text{mse}}\mathcal{L}_{\text{mse}}) + \lambda_{\text{disto}}\mathcal{L}_{\text{disto}} + \lambda_{3D}\mathcal{L}_{3D}$$

Gaussian Negative Log-Likelihood (NLL). The primary geometric loss optimizes the 1D internal coordinates via a Gaussian NLL formulation. This enables the network to aggressively penalize errors in confident regions while naturally attenuating gradients in uncertain regions:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2}e^{-\log \sigma^2}(\mu - y)^2 + \frac{1}{2}\log \sigma^2$$

Angular MSE Loss. To prevent a chiral mode collapse where the network predicts backward angular orientations but artificially minimizes the NLL loss by inflating the predicted variance, a secondary Mean Squared Error is applied to the unscaled angular unit vectors:

$$\mathcal{L}_{\text{mse}} = \text{MSE}(\mu_\theta, y_\theta) + \text{MSE}(\mu_\tau, y_\tau)$$

Distogram Loss. The distogram head is optimized using Cross-Entropy loss over 64 distance bins.

Sliding-Window 3D dRMSD Loss. A localized sliding-window Distance RMSD (dRMSD) is utilized, computing the pairwise L_1 distance error exclusively within a localized sequence window ($w = 16$):

$$\mathcal{L}_{3D} = \frac{1}{N_w} \sum_{|i-j| \leq 16} |\hat{d}_{ij} - d_{ij}|$$

This provides reasonably sized spatial gradients during the backward pass, supervising local structure without punishing the network for long-range global deviations.

D. Decoupled Physical Refinement

To mitigate compounding angular errors inherent to NeRF reconstructions, a decoupled L-BFGS optimizer [8] refines the tertiary topology during inference. The optimizer adjusts predicted angles (μ_θ, μ_τ) while holding bond distances (μ_d) rigid. Guided by the network’s epistemic uncertainty, confident secondary structures (low log-variance) are frozen by zeroing their gradients, restricting descent exclusively to uncertain hinges. The solver minimizes a bipartite objective: $\mathcal{L}_{\text{refine}} =$

$\mathcal{L}_{\text{disto_target}} + \lambda_{\text{clash}}\mathcal{L}_{\text{steric}}$. The attractive force applies an MSE against high-confidence distogram targets, while the repulsive force resolves atomic overlaps via a quadratically smoothed penalty for non-bonded C_α atoms violating a 3.8 Å collision threshold.

E. Training Procedure

Data is ingested using SideChainNet (CASP-12 dataset) [9]. To prevent sequence memorization, a 30% thinning factor is applied to strictly limit sequence identity between the training and test sets. The resulting training set consists of 24,789 proteins. Sequences with a length over 200 residues are cropped using a random sliding window, capping the $O(L^2)$ memory footprint of the pairwise matrices while preserving sufficient context for multi-domain interactions.

To maximize hardware throughput on highly variable protein lengths, a Dynamic Token Batching strategy is implemented. Proteins of similar lengths are grouped until a strict token limit is reached (e.g., 4096 tokens per batch), minimizing wasteful tensor padding. Gradient Accumulation is employed to average gradients across 20,000 tokens per step. The network is optimized using AdamW [10] with a linear warmup and cosine decay learning rate schedule.

F. Evaluation Methodology

Predicted 3D coordinates were aligned to ground-truth targets via the Kabsch algorithm [11]. Global topology was evaluated using TM-Score [12], GDT-TS [13], and RMSD. TM-Score was prioritized as the primary indicator of global accuracy because it is length-normalized and highly robust against outlier deviations such as highly flexible terminal tails, which can artificially inflate absolute RMSD. Localized secondary structure stability was measured using a per secondary structure-segment dRMSD to isolate rigid folding geometry from global orientation errors. Additionally, structural viability is audited by calculating the frequency and severity of physically impossible atomic overlaps via a steric clash penalty thresholded at 3.8 Å. To prevent length bias, all metrics were stratified into short (<200), medium (200–499), and long (≥ 500) sequence categories.

IV. RESULTS AND DISCUSSION

A. Experimental Parameters

The model was trained over 5 hours on an RX 9070 XT with 16GB of VRAM, implemented in PyTorch, and initialized with a random seed of 42. The 1D Transformer utilizes 6 layers, 8 attention heads, a feed-forward dimension of 1024, and a dropout rate of 0.1. The 2D projection heads operate with a hidden dimension of 128. The network was optimized using AdamW with a peak learning rate of 5×10^{-4} and a minimum learning rate ratio of 0.01, following a 1,000-step linear warmup and a 10,000-step cosine decay schedule. Training batches contained 2,500

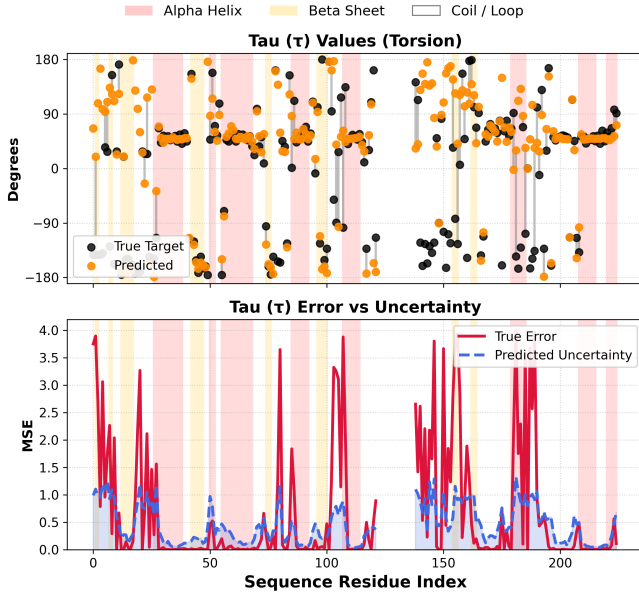


FIG. 2. Torsion angle prediction, ground truth, and error per residue. Helices are structurally correct, while sheets and coils suffer larger error and frequent chirality errors of 180 degrees. Model confidence aligns exceptionally well with error magnitude and frequency.

tokens across 8 gradient accumulation steps. Loss components were balanced using $\lambda_{\text{geom}} = 0.1$, $\lambda_{\text{disto}} = 0.2$, and $\lambda_{3D} = 1.0$. For downstream refinement, the L-BFGS optimizer applied a learning rate of 1.0 over 5 outer steps (max 20 iterations each) using a strong Wolfe line search, halting early if total loss fell below 0.1. Gradients were frozen for confident bond angles ($\log \sigma_{\theta}^2 < -1.0$) and torsions ($\log \sigma_{\tau}^2 < -2.0$). Refinement was bounded by a distogram contact cutoff of 16.0 Å, an entropy cutoff of 4.0, and a 3.8 Å steric clash threshold (excluding immediate sequence neighbors where $|i - j| \leq 2$) governed by a clash weight of $\lambda_{\text{clash}} = 10.0$.

B. Results

TABLE II. **Evaluation Metrics (Base $\rightarrow$ Refined).** Performance on a length-stratified subset of the CASP12 test split, showing the transition from unrefined 3D base coordinates to optimizer refined coordinates. Arrows indicate if higher ($\uparrow$) or lower ($\downarrow$) is better. Global N=984, Short N=603, Medium N=335, Long N=46.

Metric	Global	Short	Medium	Long
TM-Score ($\uparrow$)	0.18 $\rightarrow$ 0.31	0.20 $\rightarrow$ 0.33	0.15 $\rightarrow$ 0.30	0.08 $\rightarrow$ 0.16
GDT-TS ($\uparrow$)	0.12 $\rightarrow$ 0.22	0.18 $\rightarrow$ 0.29	0.03 $\rightarrow$ 0.11	0.00 $\rightarrow$ 0.01
RMSD (Å, $\downarrow$)	18.6 $\rightarrow$ 12.8	12.3 $\rightarrow$ 9.3	25.5 $\rightarrow$ 16.3	50.9 $\rightarrow$ 33.3
Helix dRMSD (Å, $\downarrow$)	0.56 $\rightarrow$ 0.68	0.44 $\rightarrow$ 0.51	0.65 $\rightarrow$ 0.86	1.25 $\rightarrow$ 1.31
Sheet dRMSD (Å, $\downarrow$)	0.46 $\rightarrow$ 0.57	0.36 $\rightarrow$ 0.47	0.50 $\rightarrow$ 0.62	1.05 $\rightarrow$ 1.10
Clash Severity ($\downarrow$)	0.19 $\rightarrow$ 0.25	0.16 $\rightarrow$ 0.11	0.25 $\rightarrow$ 0.47	0.22 $\rightarrow$ 0.48

The structural accuracy of the predictions varies greatly. For short to medium traces ($L \leq 500$), the TM-Score can range from 0.8 (almost perfect) to 0.1 (basically random). Generally, a TM-Score above 0.5 indicates accurate topology. As detailed in Tab. II, the mean TM-

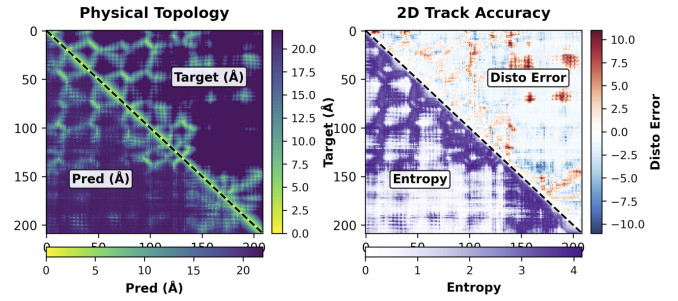


FIG. 3. Comparison between the true distogram and the predicted distogram of a good sample. This particular sample from the validation set had a base TM-Score of 0.25, while the refined topology scores a 0.73.

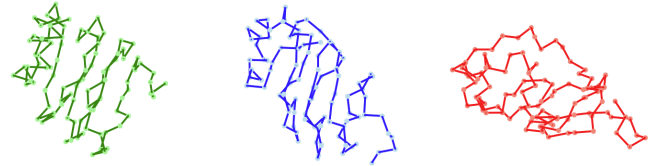


FIG. 4. A comparison of predictions. Left: Refined, Middle: True, Right: Base. In this sample, the architecture provided a particularly accurate reconstruction due to an accurate distogram prediction and reasonable base prediction.

Score is 0.31, dropping down to 0.16 for long proteins. As opposed to GDT-TS and RMSD, both of which scale negatively with length, the TM-Score is length invariant. Yet, longer proteins will always leave greater room for topological error, especially when using angle prediction, where a single error can throw off the entire topology via the lever-arm effect. Additionally, the training method used, where long proteins were cropped, may introduce a length bias in the model, even when using RoPE.

While bond angle and separation distance predictions show a small average error, the same cannot be said for torsions. Torsions are by far the most important determinant for tertiary structure, and unfortunately, the geometry head suffers from severe chiral confusion. An MSE of 4.0 is the maximum possible error of the torsion angle, representing a complete directional flip. As seen in Fig. 2, the occurrence of chiral errors is very frequent.

The geometry head provides a reliable uncertainty estimate, as its predicted variance correlates almost perfectly with absolute error. This is exactly what makes the difference between the base and refined predictions so large; these chiral errors are often entirely mitigated by the refinement process since the model assigns low confidence to the geometry head in those specific areas.

As shown in Fig. 3, even strong distogram predictions display occasional errors of 10 Å or more. Actual contact predictions are less confident, with only core contact sections displaying high certainty surrounded by an entropic blur. Despite this, the trace of 200 residues starts at a TM-Score of 0.25, increasing to an excellent 0.73 with refinement. This improvement stems entirely from the accurate distogram and the reliable expression of uncer-

tainty in the geometry head. See Fig. 4 for a comparison of base vs refined topology.

In the lower section of Tab. II, an opposite trend between base and refined results is observed. Secondary structure dRMSD as well as Steric Clash Severity increase. For secondary structures, this is due to the excellent baseline of sub-Ångström accuracy provided by the geometry head. When the optimizer is allowed to move even a small part of a helix or sheet, this near-perfect structure will naturally become slightly worse. As for steric clashes, this is largely due to base predictions being unreasonably spaced out, often taking winding paths away from the protein core, whereas real proteins are generally quite dense.

V. CONCLUSIONS AND OUTLOOK

In this project, a lightweight, single-sequence Transformer architecture for coarse-grained protein structure prediction was introduced. By using parallel tracks that only communicate at the last moment, the model remains highly computationally efficient.

The results demonstrate that this architecture successfully captures global topological scaffolding. Crucially, the integration of a Gaussian Geometry Head proved highly effective not just for coordinate prediction, but as a bridge to downstream physical solvers. By accurately correlating predicted log-variance with true angular error, the network’s epistemic uncertainty allowed a decoupled L-BFGS optimizer to surgically repair chiral traps and hinge errors, substantially improving the final TM-Score.

Moving forward, several avenues exist to address the current architectural limitations:

Resolving Chiral Confusion: While the addition of a small pure MSE loss component mitigated some chiral collapse, the 1D geometry head still struggles with 180-degree torsion flips. Future iterations could explore direct 3D coordinate attention mechanisms or predicting local backbone frames (similar to FAPE) to enforce strict 3D physical constraints natively during the forward pass.

Length Bias Mitigation: The performance degradation on sequences exceeding 500 residues suggests that the measures taken to make the model length-invariant were not enough, and more advanced RoPE augmentations may be a possible solution.

Complexity Impact Analysis: Many architectural decisions were made based on quick ablation experiments on short training runs. For example, the impact of the spatial attention pooling method used must be properly evaluated against simpler, more interpretable methods with the goal of decreasing model complexity.

VI. CONTRIBUTIONS

This report, its figures, and the code were created by Noah Andreasson.

VII. CONFLICT OF INTEREST

No conflict of interest has been identified.

VIII. DATA AND CODE AVAILABILITY

Code available upon request.

SidechainNet: <https://github.com/jonathanking/sidechainnet>

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